MIL-C-24643/31B
3 October 1986
SUPERSEDING
MIL-C-0024643/31A(SH)
8 November 1985
MIL-C-24643/31
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SU
(INCLUDING VARIATIONS LS2SA AND LS2SUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be supplied in three variations: LS2SU (unarmored), LS2SA (armored) and LS2SUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, in accordance with ASTM B 286, size 22-7, uncoated. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness of 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form each pair.
Fifth - Binder tape over each pair, at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of AWG no. 34 or no. 36 tin coated copper strands. Braid angle 20 to 40 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.

(B) denotes changes.

AMSC N/A FSC 6145
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Eighth - The required number of pairs (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. Fillers shall be used, as necessary, to form a round cable.

Ninth - Binder tape applied helically with overlap.

Construction variant for type LS2SU

Tenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LS2SA

Tenth - Cross-linked polyolefin jacket. (See table I for thickness).

Eleventh - Braided metal armor and paint.

Construction variant for type LS2SUS

Tenth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

Eleventh - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

Thirteenth - Binder tape applied helically with overlap.

Fourteenth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

TABLE I. Details.

Military 1/ part no. M24643/31	Size 2/	Number of pairs	Cable jacket thickness inch (nominal)	Overall diameter maximum (inches)					
				Type LS2SU		Type LS2SA		Type LS2SUS	
				min	max	min	max	min	max
-01UN	3	3	0.040	0.480	0.520	0.530	0.570	0.540	0.580
-02UN	7	7	.040	.610	.660	.660	.710	.670	.720
-03UN	10	10	.040	.770	.830	.820	.880	.830	.890
-04UN	14	14	.050	.860	.930	.910	.980	.920	.990
-05UN	19	19	.050	.970	1.040	1.020	1.090	1.030	1.100
-06UN	24	24	.050	1.120	1.210	1.170	1.260	1.180	1.270
-07UN	30	30	.050	1.190	1.280	1.240	1.330	1.250	1.340
-08UN	37	37	.050	1.290	1.380	1.340	1.430	1.350	1.440
-09UN	44	44	.050	1.460	1.550	1.510	1.600	1.520	1.610
-10UN	61	61	.060	1.660	1.740	1.710	1.790	1.720	1.800

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.

2/ Type LS2SU, LS2SA and LS2SUS.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.37
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield (pair shields only).....	500
Insulation resistance - shield to shield (LS2SUS only, overall shields only), volts, minimum.....	6.0
Megohms/1000 feet, minimum.....	1.0
Conductor to conductor - megohms/1000 feet, minimum	500
Jacket flaws.....	No failure
Conductor and shield (LS2SUS only) continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Capacitance	
Mutual capacitance - at 1 MHz, pF/ft, maximum.....	30
Capacitance unbalance - percent, maximum.....	8
Characteristic impedance - at 1 MHz, ohms.....	75 ± 5
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Attenuation - at 3 MHz, decibels per 100 feet, maximum..	3
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50

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EXAMINATION AND TESTS: (Continued)

	Requirements
Shrinkage	No failure
Permanence of printing (jacket) - (LS2SU and LS2SUS only) - cycles, minimum	250
Shield conformance to material, construction and coverage	No failure
Armor (LS2SA only) conformance to material, construction and coverage	No failure
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
(B) Surface transfer impedance (type LS2SUS only)	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 30	1000
37 through 61	500

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-31)

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